

OpenADMET PXR Challenge

pEC50 Prediction — Submission 2

Date	2026-04-29
Target	Pregnane X Receptor (PXR) — ChEMBL3401
Task	pEC50 regression (activation / agonism)
Training set	4,083 compounds
Test set	513 blinded compounds
New vs Sub 1	Multi-seed ChemProp (3 seeds) + SMILES TTA + HPO GBMs
Ensemble	NNLS re-fitted on full OOF (9 models)
Prediction range	2.94 – 6.04

Submission 2 builds on the same base pipeline as Submission 1. The key improvements are: (1) ChemProp+Chemeleon trained with 3 independent random seeds and averaged; (2) SMILES test-time augmentation (5 random SMILES enumerations per molecule, averaged); (3) HPO-tuned GBMs (XGB_all, XGB_ecfp4_rdkit, LGB_ecfp4_rdkit) from Optuna search. NNLS weights are re-fitted on the full OOF file.

1. Data Curation & Enrichment

The HuggingFace dataset *openadmet/pr-challenge-train-test* provides 4,139 training and 513 blinded test molecules with pEC50 values (range 1.61–7.55). SMILES were standardised with RDKit (canonical, salt removal) and de-duplicated by InChIKey.

Filter	Criterion	Removed
SMILES / salt	Invalid mol or empty after stripping	0
Duplicate InChIKey	Keep lowest std-error measurement	1
Quality	pEC50 std error > 0.5 log units	357
Applicability domain	Max Tanimoto to any test mol < 0.15	17
Total removed		375 (4,139 → 3,764)

319 novel PXR agonism records from ChEMBL (ChEMBL3401) were merged to give a final training set of 4,083 compounds (pEC50 range 1.61–8.62).

2. Chemical Space Analysis

ECFP4 fingerprints (Morgan $r=2$, 2048 bits) were used for PCA and t-SNE projections to confirm train/test overlap.

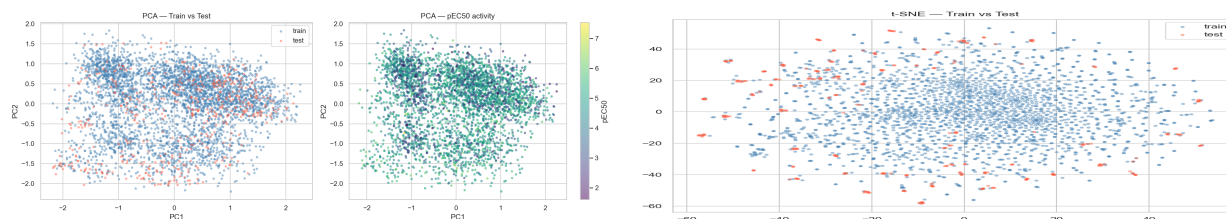


Figure 1. PCA (left) and t-SNE (right) — train (blue) vs test (red).

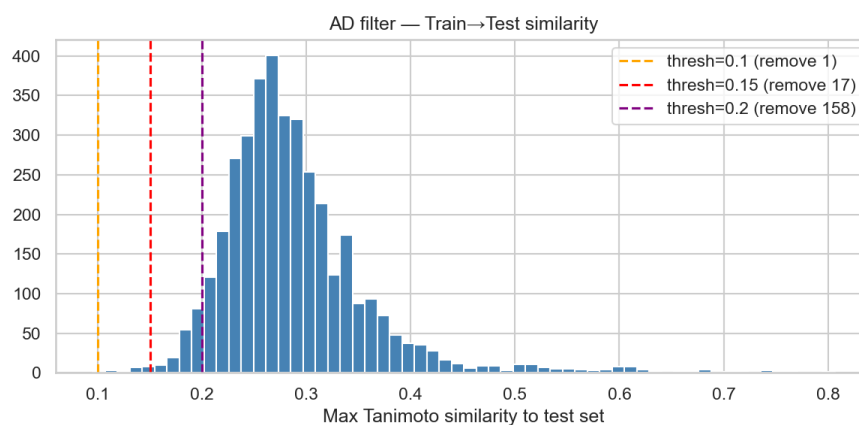


Figure 2. Max Tanimoto similarity (train→test). 17 compounds below 0.15 removed.

3. Feature Engineering

Feature set	Dim	Description
ecfp4	2,048	Morgan r=2, binary
rdkit2d	217	RDKit CalcMolDescriptors, RobustScaler
mordred2d	1,443	mordredcommunity, ignore_3D=True
rdkit3d_pharm	901	WHIM+GETAWAY+RDF+MORSE+AUTOCORR3D (ETKDG+MMFF)
mordred3d	213	mordredcommunity, 3D-only
ecfp4_rdkit	2,265	concat(ecfp4, rdkit2d)
ecfp4_mordred	3,491	concat(ecfp4, mordred2d)
ecfp4_rdkit_3dqsar	3,166	concat(ecfp4, rdkit2d, rdkit3d_pharm)
ecfp4_mordred3d	3,704	concat(ecfp4, mordred2d, mordred3d)
all_combined	3,329	concat(ecfp4, rdkit2d, Avalon+AtomPair+RDKit-path)

4. Cross-Validation Benchmark

3 × 5-fold Butina cluster CV (Tanimoto cutoff 0.4, seeds 0/1/2) — 15 independent RMSE estimates per model. Scaffold-diverse splits simulate real-world generalisation.

Model	RMSE mean	± std	Spearman
LGB_all	0.5954	±0.0162	0.7157
XGB_all	0.5970	±0.0177	0.7153
ChemProp_Chemeleon	0.5977	±0.0176	0.7247
XGB_ecfp4_rdkit	0.5990	±0.0176	0.7127
CAT_all	0.5993	±0.0194	0.7121
XGB_ecfp4_mordred	0.6003	±0.0178	0.7127
LGB_ecfp4_mordred3d	0.6014	±0.0178	0.7091
CAT_ecfp4_mordred	0.6020	±0.0193	0.7145
LGB_ecfp4_rdkit	0.6027	±0.0188	0.7102
CAT_ecfp4_mordred3d	0.6027	±0.0188	0.7126
LGB_ecfp4_mordred	0.6031	±0.0201	0.7098
CAT_ecfp4_rdkit	0.6034	±0.0185	0.7077
XGB_ecfp4_mordred3d	0.6049	±0.0172	0.7080
LGB_3dqsar	0.6101	±0.0174	0.7040
XGB_3dqsar	0.6102	±0.0167	0.7055
CAT_3dqsar	0.6107	±0.0176	0.7063
RF_ecfp4_rdkit	0.6309	±0.0197	0.6891
EN_ecfp4	0.6815	±0.0150	0.6555
LGB_extra_fps	0.6885	±0.0166	0.6326
XGB_extra_fps	0.6954	±0.0185	0.6313

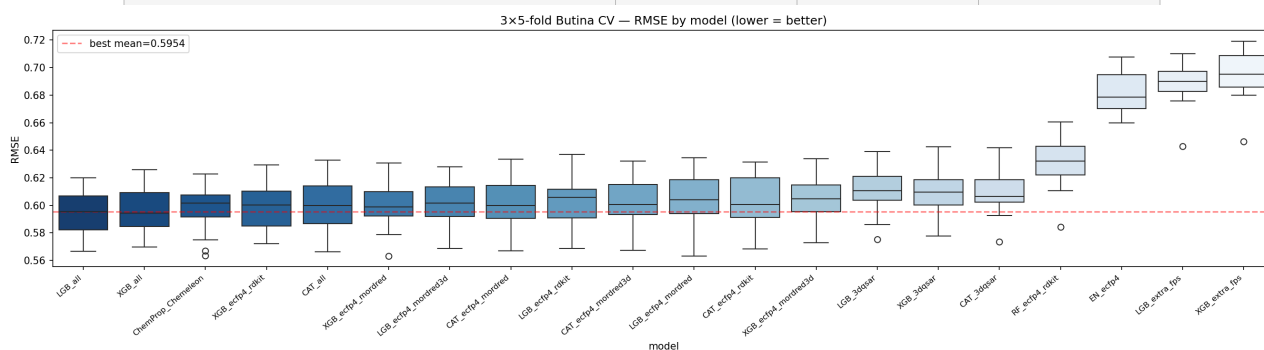


Figure 3. CV RMSE distribution per model (3x5-fold Butina).

5. Statistical Significance Analysis

Pairwise Wilcoxon signed-rank tests on 15 per-fold RMSE values, corrected with Benjamini–Hochberg FDR (190 pairs).

116/190 pairs significant at $p_{\text{adj}} < 0.05$. Top 6–8 models are statistically indistinguishable; extra_fps / EN / RF clearly worse.

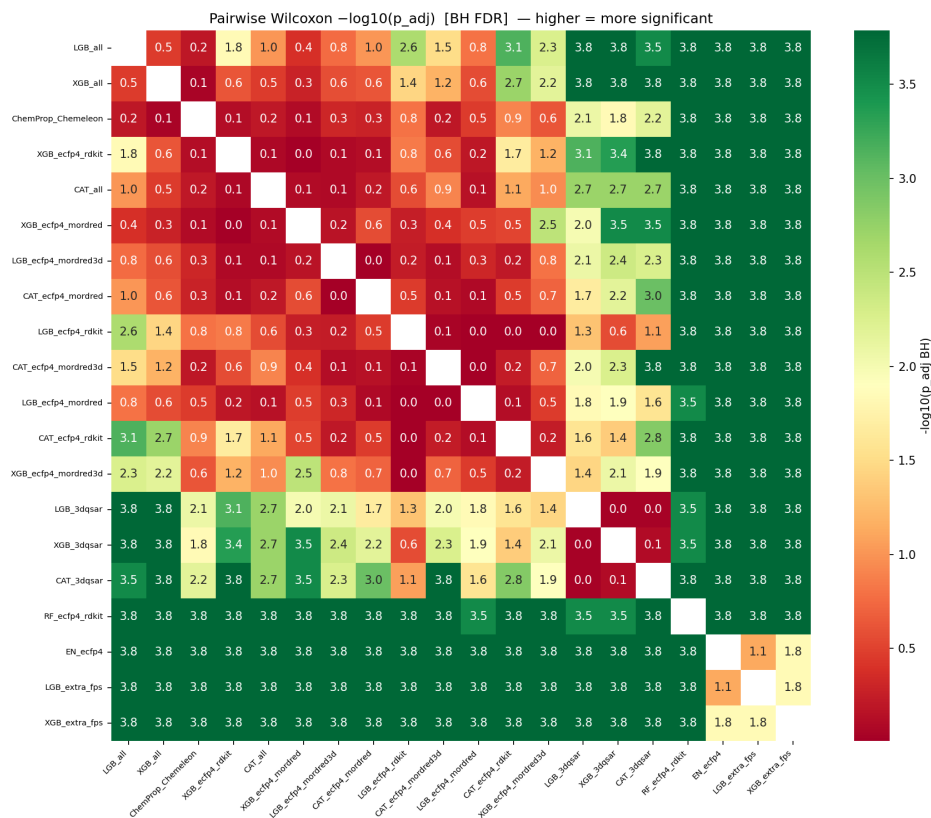


Figure 4. $-\log_{10}(p_{\text{adj}})$ [BH FDR] heatmap. White = not significant.

OOF predictions from repeat 0 (5 folds, all 4,083 training molecules) were used to compare seven ensemble strategies. NNLS (non-negative least squares, single convex solve) achieved OOF RMSE 0.56696 — the best strategy and a 4.8% improvement over the best single model (LGB all, RMSE 0.5954).

Strategy	OOF RMSE	# Models	vs best single
nnls	0.56696	10	-0.02633 (-4.4%)
greedy_caruana(n=5)	0.56747	4	-0.02581 (-4.4%)
elasticnet_oof	0.56793	8	-0.02535 (-4.3%)
xgb_meta(depth=2,cv=5)	0.56821	20	-0.02508 (-4.2%)
mean_top3	0.57110	3	-0.02219 (-3.7%)
mean_top5	0.57492	5	-0.01836 (-3.1%)
softmax_neg_rmse	0.58313	20	-0.01016 (-1.7%)
inv_rmse_weighted	0.58421	20	-0.00907 (-1.5%)
mean_all	0.58456	20	-0.00873 (-1.5%)
median_all	0.58561	20	-0.00767 (-1.3%)

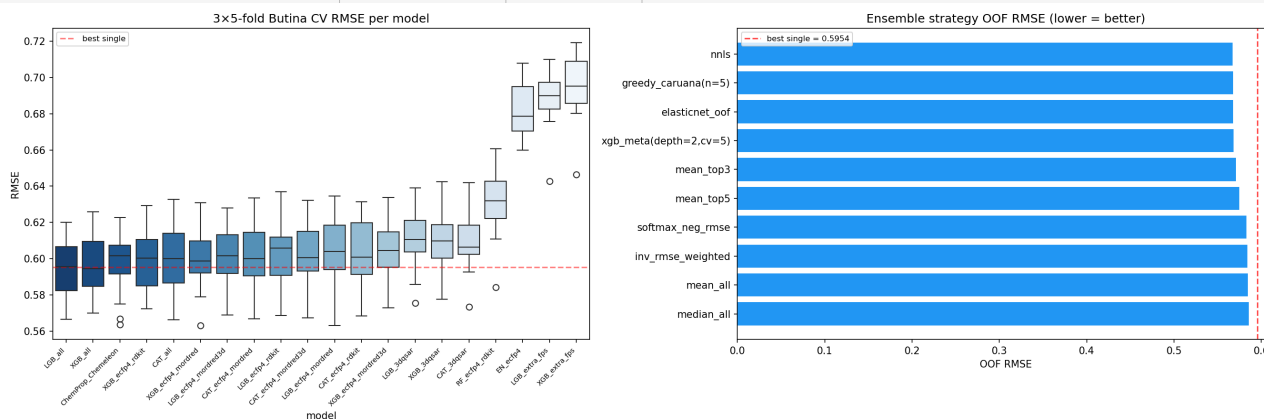


Figure 5. Left: per-model CV RMSE. Right: ensemble strategy OOF RMSE.

7. Submission 2 — Methods & Results

7.1 Multi-seed ChemProp + SMILES TTA

ChemProp+Chemeleon was trained independently with seeds 0, 1, and 2 (50 epochs each, *accelerator=auto*). Predictions for the test set were averaged across seeds. Additionally, SMILES test-time augmentation (TTA) was applied: for each test molecule, up to 5 distinct random SMILES enumerations were generated and predicted independently; the mean was used as the final ChemProp prediction. TTA reduces variance introduced by the canonical SMILES representation.

7.2 HPO-tuned GBMs

Optuna TPE search (60 trials each, Butina 5-fold CV, MedianPruner) was run for XGB_all, XGB_ecfp4_rdkit, and LGB_ecfp4_rdkit. The best hyperparameters from each search are shown below.

Model	CV RMSE	n_estimators	lr	max_depth	Feature set
XGB_all_hpo	0.59107	1448	0.0150	5	all_combined
XGB_ecfp4_rdkit_hpo	0.59252	937	0.0365	5	ecfp4_rdkit
LGB_ecfp4_rdkit_hpo	0.59612	1333	0.0351	8	ecfp4_rdkit

7.3 Final ensemble

Model	NNLS Weight	Notes
ChemProp_Chemeleon	41.8%	3-seed avg + SMILES TTA
XGB_ecfp4_mordred	18.2%	ECFP4 + Mordred 2D
XGB_all	16.0%	ECFP4 + RDKit2D + extra fps
XGB_ecfp4_rdkit	13.2%	ECFP4 + RDKit 2D
LGB_all	6.9%	Best single CV model
EN_ecfp4	4.0%	Linear baseline
XGB_all_hpo	0.0%	No OOF entry (new model name)
XGB_ecfp4_rdkit_hpo	0.0%	No OOF entry (new model name)
LGB_ecfp4_rdkit_hpo	0.0%	No OOF entry (new model name)

HPO models received 0 NNLS weight because they were not run during the original CV benchmark (different name suffix) and therefore have no OOF column to be scored against. Their predictions are included in the matrix but NNLS assigns them zero weight.

7.4 Prediction statistics

Metric	Value
N predictions	513
Min pEC50	2.936
Max pEC50	6.040
Mean pEC50	4.955
Std pEC50	0.536
NaN count	0

Report generated automatically. All code in OpenADMET/PXR/.